

Review - 2 Report Group - 14

A general anomaly detection framework for fleet-based condition monitoring of machines

1. Given a hypothetical situation where more than 2/3 of the machines are faulty, how will your architecture handle it?

The paper we choose works on the assumption that majority of the machines are healthy, so the algorithms given in the paper can't be used directly to handle such a situation. But, the CI models we are integrating into the edge layer will be able to find the faulty machines even under the given constraints. So, using the CI models will be the best solution.

2. What algorithm is used in unsupervised learning?

Hierarchical clustering method is used in the unsupervised learning.

3. The fuzzy model also does a hard clustering, how is this different from the unsupervised learning model?

The unsupervised model clusters the machines in a binary fashion, ie, either faulty or not faulty. The fuzzy models doesn't cluster the machines as black or white, rather it clusters them in shades of grey, like faulty, approaching faulty and healthy. This makes sure all the machines are clustered properly and each cluster can be processed more efficiently.